

Faculty of Information Technology (FIT) – Ho Chi Minh City University of Science (HCMUS)

CS423 / CSC13003 – Software Testing (AI-augmented · 2026)

1. Student information

- **Student name (printed):** Trần Trí Nhân
- **Student ID:** 23127097
- **Class / Cohort:** Kiểm thử phần mềm - 23KTPM1
- **Assignment ID:** HW03
- **Assignment date:** 26/07/2026
- **AI tool(s) used:** Claude
- **AI tool(s) used:**
 - ☒ Yes
 - ☐ No

2. Audit Table

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning	(5) Student Fix
Tool: Claude Time: 15:17 25/07/2026 Input: 4. System Under Test (SUT) SUT: EMS — Event Management System, Faculty of Information Technology. A web application for creating, publishing, and running academic events, with administration, participant registration, check-in, support requests, analytics, and system configuration. Web (SUT): https://promoter-starboard-prude.ngrok-	# EMS GUI Checklist Item ID Item Description IA Mapping Heuristic/Principle Mapping Result Notes --- --- --- --- --- GUI-001 All text on a page uses a consistent font family (no mixed/inconsistent fonts within one view) IA-01 Nielsen: Consistency & Standards; Shneiderman: Strive for Consistency GUI-002 Primary action buttons (e.g., Save, Publish, Register) use the same color across the app IA-01 Nielsen: Consistency & Standards	INCOMPLETE	While the AI correctly followed the principles and the IA requirements to produce a GUI checklist, it still missed some items. For the manually added item of IA-01, since the AI had to cover many requirements of IA-01, once it already covered the color(the	The team added 3 items, one for IA-01, one for IA-02 and one IA-03.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning	(5) Student Fix
free.dev/ Admin account (for admin scenarios A and C, and the admin side of D): admin@gmail.com / Admin@123 — the account must hold the ADMIN role on EMS. User accounts (for the user side of scenarios B and D): register your own student / lecturer / guest account through the EMS sign-up flow. Do not share a single account across the group for the user-side scenarios — each member needs their own so their actions are distinguishable. The application is served through an ngrok tunnel and its data may be reset periodically. Take your evidence (screenshots, recordings) as you go; do not assume a state you created earlier will still be there next session. The EMS features are organised into the following pools, which map onto the four scenarios in §5: * Pool A — Event administration. Dashboard KPIs (Total Events, Total Check-ins, Attendance Rate, Total Users); Events list; Add/Edit Event (thumbnail 4:3 + banner 24:9 upload, Rich-Text	GUI-003 Visible page elements are grid-aligned with no overlapping or clipped components IA-01 Nielsen: Aesthetic & Minimalist Design; Norman: Constraints GUI-004 Switching the language toggle (EN/VI) translates all visible text on the page with no leftover untranslated strings IA-01 Nielsen: Match Between System & Real World GUI-005 A screen with zero records shows an explicit empty-state message IA-01 Nielsen: Help Users Recognize/Diagnose; Norman: Feedback GUI-006 A screen fetching data shows a loading indicator (spinner/skeleton) before content renders IA-01 Nielsen: Visibility of System Status; Norman: Feedback GUI-007 Foreground text has sufficient contrast against its background to be legible IA-01 Nielsen: Aesthetic & Minimalist Design GUI-008 Every icon-only control has an accompanying text label or tooltip identifying its function IA-01 Nielsen: Recognition Rather Than Recall GUI-009 Date/time values are displayed in one consistent format throughout a given page IA-01 Nielsen: Consistency & Standards		primary buttons), it moved on without paying attention to the text color. For the manually added item of IA-02, the AI missed this item simply because it did not know about the UI implementation of EMS beforehand, so instead it chose a setting time item to avoid incorrect items. For the manually added item of IA-03, since the prompt included the description of IA-03, the AI based its result exactly on that, without realizing IA-03 was actually about navigation, therefore it missed this item about navigating with Tab key.	

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning	(5) Student Fix
content, date/time validation); registration configuration (student/lecturer/guest toggles, Max Slots, Waitlist, additional roles); Draft / Publish / Preview / Important Update / Delete; Participants & Reviews approval; Check-in. * Pool B — Participant experience. Public home with the featured-event carousel; category browsing and search; event detail; registration form (role selection, waitlist); My Registrations and the barcode/QR ticket; post-event star reviews. * Pool C — User administration. Users list (Avatar+Name, Role, Member Code, Active, Audit columns); Assign Role; Block / Unblock; Reset Password; Export to Excel; audit log. * Pool D — Support requests. User side: create a support request (category, content, image attachment), My Requests list and detail with the official response. Admin side: Support Requests list (Pending / Resolved tabs, search by member code or category), request detail with image lightbox, internal note,	GUI-010 The same status value (e.g., "Active," "Pending") uses the same badge color everywhere it appears IA-01 Nielsen: Consistency & Standards; Norman: Mapping GUI-011 Every page displays a title/header identifying its content IA-01 Nielsen: Match Between System & Real World GUI-012 No placeholder/debug text (e.g., "Lorem ipsum," "TODO") is visible anywhere in the UI IA-01 Nielsen: Aesthetic & Minimalist Design GUI-013 Every required form field displays a visible required-indicator (e.g., asterisk) IA-02 Nielsen: Error Prevention; Shneiderman: Offer Simple Error Handling GUI-014 A field's label stays visible/associated with its input after the user enters a value IA-02 Nielsen: Recognition Rather Than Recall GUI-015 A validation error message appears directly next to the specific field it concerns IA-02 Nielsen: Help Users Recognize/Diagnose/Recover from Errors GUI-016 A validation error message states what is wrong and how to correct it (not a generic "Invalid") IA-02 Nielsen: Help Users			

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning	(5) Student Fix
and official response. Beyond the functional pools above, this homework focuses on the user interface. The interface concerns are organised into four interface aspects (IA) — used as the coverage dimensions of the shared checklist: * IA-01: General UI standards (layout, alignment, typography, colour, consistency, i18n EN/VI, empty/loading states). * IA-02: Forms (labels, validation, error placement, required-field handling, uploads, rich-text editor). * IA-03: Navigation (menus, breadcrumbs, tabs, sidebar, drag-and-drop reorder, back/return actions, deep links). * IA-04: Feedback / state (toasts, badges, confirmation dialogs, progress bars, status colours, real-time updates). Read the above description carefully to understand EMS description and the IA-0x requirements. Next, use nielsen's usability heuristics, norman principles and Shneiderman golden	Recognize/Diagnose/Recover from Errors GUI-017 Submitting a form with an empty required field is blocked and shows an inline error instead of submitting IA-02 Nielsen: Error Prevention GUI-018 Selecting an event end date/time earlier than the start date/time is rejected with a clear error IA-02 Nielsen: Error Prevention GUI-019 The accepted file type/size for an image upload is stated before the user selects a file IA-02 Nielsen: Help & Documentation; Norman: Signifiers GUI-020 Selecting an image for upload shows a visual preview before the form is submitted IA-02 Norman: Feedback GUI-021 The declared image aspect ratio (4:3 thumbnail / 24:9 banner) is enforced or a mismatch warning is shown IA-02 Nielsen: Error Prevention GUI-022 The Rich-Text editor toolbar visually reflects the formatting state (e.g., Bold icon highlighted) at the cursor position IA-02 Norman: Feedback; Nielsen: Visibility of System Status GUI-023 Data entered in a form is retained (not cleared) after a failed submission IA-02 Nielsen: User Control & Freedom			

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning	(5) Student Fix
rules (look these up on google for the latest version), create a GUI checklist that strictly follow the following requirements: * > 40 items total * must cover all four interface aspects IA-01... IA-04. * Each item must be atomic (one thing) and binary-testable(Pass/Fail unambiguously) * Each item must be testable regardless of the screen(independent of screen) The GUI checklist must be executable under these following scenarios and screens: Scenarios: - Scenario A — Admin creates and manages events. Function group: the event lifecycle on the admin side. - Scenario B — User registers to attend an event. Function group: public discovery and participant registration. - Scenario C — Admin manages users. Function group: user administration. - Scenario D — User requests Support and Admin resolves it. Function group: the	GUI-024 A disabled submit button is visually distinct from an enabled submit button IA-02 Norman: Constraints/Signifiers GUI-025 The sidebar/menu item for the current section is visually highlighted as active IA-03 Nielsen: Visibility of System Status GUI-026 The breadcrumb trail (where present) accurately reflects the current page's position in the hierarchy IA-03 Nielsen: Match Between System & Real World GUI-027 Using the Back/Return action returns the user to the previous screen with no unintended data loss IA-03 Nielsen: User Control & Freedom; Shneiderman: Support Internal Locus of Control GUI-028 The currently selected tab (e.g., Pending/Resolved) is visually distinguished from unselected tabs IA-03 Nielsen: Visibility of System Status GUI-029 Switching tabs replaces the content area fully, with no residual content from the previous tab IA-03 Nielsen: Consistency & Standards GUI-030 Opening a direct URL (deep link) to a detail record loads that exact record's data IA-03 Nielsen: Match Between			

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning	(5) Student Fix
support-request lifecycle across both the user and admin sides. - Scenario A: (A1) Events list with status filters and notification dots; (A2) Add/Edit Event form — image upload + Rich-Text + date/time validation; (A3) Registration & Roles configuration panel — Max Slots / Waitlist / additional role; (A4) Participants & Reviews approval — status colours, progress bar, Export; (A5) Check-in tab — scan-state handling and real-time log. - Scenario B: (B1) Home / events listing — featured carousel, categories, search/filter; (B2) Event detail page — banner, schedule, register button, waitlist notice; (B3) Registration form — role selection, additional role, confirmation; (B4) My Registrations / ticket — status and barcode/QR; (B5) Post-event review — 1–5 star rating. - Scenario C: (C1) Users list — search, role/active filters, columns; (C2) Assign Role / edit user; (C3) Block-Unblock and Reset-Password dialogs — confirmation + audit; (C4) Export to Excel — column completeness	System & Real World GUI-031 A drag-and-drop reorder handle is visually distinguishable from non-draggable elements IA-03 Norman: Affordance/Signifiers GUI-032 A drag-and-drop reorder change persists after the page is refreshed IA-03 Nielsen: User Control & Freedom GUI-033 The search input is reachable from the listing screen without an extra navigation step IA-03 Shneiderman: Enable Frequent Users to Use Shortcuts GUI-034 Filter controls are visually separated from and clearly labeled apart from search controls IA-03 Nielsen: Recognition Rather Than Recall GUI-035 Navigating away from an in-progress unsaved form triggers a confirm-discard prompt IA-03 Nielsen: Error Prevention; Shneiderman: Permit Easy Reversal of Actions GUI-036 Pagination controls display the current page number and total pages/records IA-03 Nielsen: Visibility of System Status GUI-037 A successful action (e.g., save, publish, register) triggers a confirming toast/notification IA-04 Nielsen: Visibility of			

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning	(5) Student Fix
and download feedback. - Scenario D: (D1) User — create support request form with image attachment; (D2) User — My Requests list and detail with the response; (D3) Admin — Support Requests list, Pending/Resolved tabs, search; (D4) Admin — request detail — image lightbox, internal note, official response. The final GUI checklist must include these columns: item id(format as GUI-xxx), item description, IA-0x mapping, heuristic/principle mapping, result, notes. The last two columns are empty and will be filled after the actual runs. Return the final GUI checklist in markdown format. No further explanation.	System Status; Shneiderman: Offer Informative Feedback GUI-038 A failed action triggers a toast/notification clearly stating the failure IA-04 Nielsen: Help Users Recognize/Diagnose/Recover from Errors GUI-039 A destructive action (e.g., Delete, Block) requires confirmation via a dialog before executing IA-04 Nielsen: Error Prevention; Shneiderman: Permit Easy Reversal of Actions GUI-040 The confirmation dialog names the specific action and target it is confirming IA-04 Nielsen: Recognition Rather Than Recall GUI-041 A notification dot/badge count updates immediately after its triggering event occurs IA-04 Nielsen: Visibility of System Status GUI-042 A progress bar's displayed percentage matches the actual numeric ratio it represents (e.g., slots filled) IA-04 Norman: Feedback GUI-043 The same status value uses the same color coding consistently on every screen it appears IA-04 Nielsen: Consistency & Standards GUI-044 New real-time log entries (e.g., check-in log) appear without requiring a			

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning	(5) Student Fix
	manual page refresh IA-04 Nielsen: Visibility of System Status GUI-045 The current system state of a record (e.g., Draft/Published) is visibly indicated at all times on its screen IA-04 Nielsen: Visibility of System Status GUI-046 An administrative action (role change, password reset) produces a visible audit-log entry IA-04 Nielsen: Visibility of System Status; Shneiderman: Design Dialogs to Yield Closure GUI-047 Triggering an Export action shows visible feedback that the file is being generated/downloaded IA-04 Shneiderman: Offer Informative Feedback GUI-048 A button shows a temporary disabled/loading state during async processing to prevent duplicate submissions IA-04 Nielsen: Error Prevention			

3. Summary of AI Accuracy

Metric	Count	Percentage
Total AI-generated artifacts audited	1	
VALID (correct, accepted as-is)	0	0%
INVALID (wrong; rejected)	0	0%
INCOMPLETE (acceptable after edits)	1	100%

4. Conclusion

In conclusion, Claude did an amazing work of creating a generic GUI checklist for the first task of this homework. Prompt engineering played a key role in guiding Claude to generate a suitable checklist by giving the AI enough information about the topic and the requirements. However, despite the satisfactory result, the AI still missed some items and left some gaps on checklist that the team had to review and improved upon. Because the AI had to focus on covering many interface aspect(LA) requirements, while making sure that items are testable and independent, it simply could not cover all possible checklist items for the EMS website. For future usage, I would definitely reuse the prompting technique used in this homework to achieve better outcome from the AI.

5. Mandatory Disclosure

The GUI checklist was initially generated by Claude; I reviewed and modified the checklist, added item GUI-049; the screens selection rationale, the checklist execution, the cross-browser execution, the bug report, the bug finding logs and the test summary was written entirely by me. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category.

Signature

- **Student name (printed):** Trần Trí Nhân
- **Student ID:** 23127097
- **Class / Cohort:** Kiểm thử phần mềm - 23KTPM1
- **Course:** CS423 / CSC13003 – Software Testing
- **Instructor:** Lâm Quang Vũ, Hồ Tuấn Thanh, Trương Phước Lộc
- **Date:** 26/07/2026
- **Signature:**



Reference

- Kharbach, M. (2026). AI Use Policy Templates for Higher Education. CC BY-NC-SA 4.0.
- ISTQB Foundation Level Syllabus (latest version).
- Hardman, P. (2025). A Post-AI Learning Taxonomy.
- Fuster Rabella, M. (2025). OECD Education Working Paper No. 338.
- Perkins, M., Roe, J., & Furze, L. (2025). AI Assessment Scale.
- Anthropic (2025). Building reliable AI test agents — engineering blog.
- DeepEval & Promptfoo documentation — testing frameworks for LLM systems.